

KELETSO MAKALELA

Software Developer | Business Analyst | AI & Automation Engineer | Full Stack Developer

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GitHub: github.com/KeletsoNIT | Portfolio: keletsonit.github.io/Portfolio

PROFESSIONAL SUMMARY

Results-driven Software Developer with a BSc in Information Technology and hands-on experience building full-stack applications, backend systems, and AI-powered software. Currently an Agentic AI Software Developer Intern, designing intelligent, multi-agent systems that translate business requirements into working application features using Python, LangGraph, LLMs, and REST APIs. Demonstrated ability to take a complex business problem from requirements through to a working, production-style solution: designed and built an AI-powered Recruitment Agent capable of CV parsing, semantic candidate matching (FAISS), automated email communication, PDF report generation, and multi-agent workflow orchestration, including diagnosing and fixing a critical scoring bug in the matching algorithm. Comfortable working across the stack (Python, React, Node.js, Flask, SQL, Docker, Git) and across the business-to-engineering boundary, with strong requirements analysis, debugging, testing, and stakeholder communication skills. Seeking a Software Developer, Backend Developer, Full-Stack Developer, or AI Developer role where I can keep building reliable, well-architected software.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C#, C++, PHP, SQL, HTML5, CSS3

Frameworks & Libraries: React, Node.js, Flask

AI & Automation: LangChain, LangGraph, AI Agents, OpenAI API, Gemini API, Prompt Engineering, Machine Learning, Semantic Search (FAISS), Process & Email Automation

Backend & Data: REST API Development, Authentication, SQLite, MySQL, Database Design, Data Analysis, Statistics

Tools & DevOps: Git, GitHub, Docker, Linux, Streamlit, Visual Studio Code

Practices: Object-Oriented Programming, Agile Development, System Design, Software Testing, Debugging, Requirements Analysis, Technical Documentation

PROFESSIONAL EXPERIENCE

Agentic AI Software Developer Intern | New Island Technologies *Jan 2026 – Present*

- Analyse business requirements and translate them into functional application features and automated, multi-agent workflows using Python and LangGraph.
- Design and build AI-powered software solutions and RESTful APIs that integrate LLMs (OpenAI API, Gemini API) with databases and external systems.
- Implement prompt engineering techniques to improve AI response quality, accuracy, and reliability.
- Investigate and resolve data, database schema, integration, and application issues through structured debugging (e.g. session-state bugs, SQLite schema mismatches, scoring/matching logic errors).
- Design modular, maintainable software architectures following clean code and software engineering best practices.
- Collaborate within an Agile environment using Git and Docker, contributing to testing, validation, and delivery of production-ready AI solutions.

Software Developer Intern | Dev Shack *Jul 2025 – Nov 2025*

- Developed production-ready full-stack functionality using React.js and Node.js.
- Integrated REST APIs and authentication systems to improve application security and functionality.
- Collaborated within Agile development teams to deliver software features under project deadlines, using Git for version control and Docker to streamline development and deployment.
- Participated in debugging, testing, and continuous improvement of existing applications, contributing to scalable and maintainable software design.

IT Support Technician | Richfield Graduate Institute *Feb 2024 – Nov 2024*

- Provided first-line technical support for staff and students, diagnosing and resolving hardware, software, and networking issues.
- Identified user needs and systematically analysed technical problems to determine appropriate solutions.
- Maintained computer laboratories and institutional IT infrastructure, improving system availability through proactive troubleshooting.

- Communicated technical solutions clearly and patiently to users with varying levels of technical knowledge.

Software Engineering Tutor | Richfield Graduate Institute2023 – 2024

- Tutored students in Java, Python, C++, C#, and Software Engineering principles, including object-oriented design.
- Assisted students with debugging, assignments, and practical programming projects through one-on-one mentoring.
- Translated complex programming concepts into clear explanations while balancing tutoring against other academic and work commitments.

FEATURED TECHNICAL PROJECTS

AI Recruitment Platform — Python, Flask, LangGraph, OpenAI API, SQLite, FAISS, Docker

- Designed and built an AI-powered recruitment platform that automates candidate screening, matching, ranking, and reporting through a modular, three-agent workflow.
- Built AI agents for automated CV parsing/information extraction, semantic candidate matching (FAISS vector embeddings), and interview recommendations, improving on keyword-based matching.
- Automated interview invitation/rejection emails and PDF report generation for recruiters.
- Diagnosed and resolved a critical scoring bug (synonym expansion inflating scores, missing word boundaries, double-counted bonuses) by redesigning the matching algorithm and validating the fix.
- Built a custom “Screening Console” Streamlit UI (dark theme, radial match-signal gauge, dossier-style candidate cards) and produced a 10-slide stakeholder pitch deck.

FinancePilot — LangChain, LangGraph, Gemini API, Streamlit, SQLite

- Built a proactive cash flow and budget advisory AI agent around a five-node LangGraph pipeline: transaction sync, categorisation, forecasting, risk, and advisory chat.
- Implemented pandas/numpy-based seasonality and trend detection, paired with Gemini-generated narrative financial advice.
- Added synthetic data generation and a manual entry form to support an interactive, demo-ready portfolio experience.

Personal Developer Portfolio — React, JavaScript, Vite, Framer Motion

- Designed and built a fully responsive portfolio website showcasing projects, technical skills, and professional experience, using reusable React components and animated interactions.

EDUCATION

Bachelor of Science in Information Technology

Richfield Graduate Institute of Technology | 2022 – 2024

Academic Average: 73%

LEADERSHIP & ACHIEVEMENTS

- Developed an AI-powered Recruitment Agent demonstrating end-to-end AI, backend development, and automation skills.
- Hackathon Participant — Team Leader.
- Deputy Head Girl, Representative Council of Learners.
- Balanced full-time academic study with tutoring and IT support responsibilities.

CORE COMPETENCIES

Software Development | Backend & Full-Stack Development | Artificial Intelligence | AI Agent Development | REST API Development | Database Design | Process Automation | Requirements Analysis | Git Version Control | Docker | Agile Development | Problem Solving | Debugging & Testing | Technical Documentation | Team Collaboration & Communication